Gideon A. Afriyie

gideonafriyie23@gmail.com | gideonafriyie.github.io | gideon-afriyie

Education

Gettysburg College *Gettysburg, PA - USA*

B.A Mathematics, Minor in Economics *2023 - 2027*

- Honors: AMS–Math Alliance F-GAP Scholar; Taylor Gaw '13 Endowed Fellowship; Seibert Project Award

Experience

Advancing Science Lab *On-campus*

Research Assistant – 3D Topographic Modeling *Aug 2026 – Present*

- Developed and 3D-printed an initial terrain prototype using USGS elevation data, QGIS, TouchTerrain, and Creality Print.
- Verified geographic coverage using QGIS; currently evaluating scale, vertical exaggeration, and print quality.

Portland State University

altREU Research *Jun – Aug 2026*

- Contributed to a 4-person ML research team analyzing 10 years of EPA/NASA air-quality and environmental data.
- Co-authored a published research chapter reporting an ensemble model with test $R^2 = 0.825$ and MAE = 11.9 AQI points.

Under Canvas Inc.

Data Analytics Fellow *May – Jul 2025*

- Analyzed a $1.6M organizational compensation dataset using Python and Excel to identify anomalies across employee groups.
- Developed quantitative recommendations for incentive restructuring; findings presented to leadership.

Publications

Afriyie, Gideon A. (2026). Extreme Value Theory Analysis of Prime Gap Distributions: Statistical Analysis of Cramér's Conjecture and Light-Tailed Behavior. *Statistics & Probability Letters*. DOI: 10.1016/j.spl.2026.110898

Haque, R., Kebede, B., **Afriyie, G. A.**, & Yusuf, H. (2026). Predicting Urban Air Quality Trends Using Machine Learning and Public Satellite/Sensor Data. *Summer Proceedings 2026*, Chapter 16. Link

Projects

Quantitative Risk & Portfolio Optimization | GitHub

- Analyzed 6+ years of prices across 8 assets, estimating returns, volatility, and correlations (VOO–VIXY: -0.754).
- Optimized portfolios using mean-variance analysis and efficient frontiers, applying KKT conditions to restrict short selling.

Skills

Python, Excel, SQL, LaTeX

Research & Programs

2026 **Probability for AI**, Stanford University *Oct–Nov*

2026 **TPU Research Cloud**, Google *May*

2026 **AI Summer School**, MIT/Harvard IAIFI *August*